



Semester Two Examination, 2013

Question/Answer Booklet

STAGE 3AB CHEMISTRY

STUDENTS NAME: _____

TEACHERS NAME: _____

Marks	Actual	% Mark
Sec I : M/C	/ 50	
Sec II short answer	/ 70	
Sec III Extended	/ 80	
		200

Time allowed for this paper

Reading time before commencing work:

ten minutes

Working time for paper:

three hours

Material required/recommended for this paper

To be provided by the supervisor

This Question/Answer booklet

Multiple-choice Answer sheet

Chemistry Data sheet

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils, sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: non-programmable scientific calculators approved for use in the WACE examinations

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to Be answered	Suggested working time (minutes)	Marks available	Percentage of exam
Section One: Multiple-choice	25	25	50	25	25
Section Two: Short answer	8	8	55	70	35
Section Three: Extended answer	6	6	70	80	40
				Total	100

Instructions to candidates

1. The rules for the conduct of Western Australian external examinations are detailed in the *Year 12 Information Handbook 2013*. Sitting this examination implies that you agree to abide by these rules.
2. Answer the questions according to the following instructions.

Section One: Answer all questions on the separate Multiple-choice Answer sheet provided. For each question shade the box to indicate your answer. Use only a **blue** or **blackpen** to shade the boxes. If you make a mistake, place a cross through that square then shade your new answer. Do not erase or use correction fluid/tape. Marks will not be deducted for incorrect answers. No marks will be given if more than one answer is completed for any question.

Sections Two and Three: Write your answers in this Question/Answer Booklet.

3. When calculating numerical answers, show your working or reasoning clearly. Express numerical answers to **three significant figures and include appropriate units where applicable**.
4. You must be careful to confine your responses to the specific questions asked and to follow any instructions that are specific to a particular question.
5. Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.
 - Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
 - Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of question(s) that you are continuing to answer at the top of the page.
6. The Chemistry Data Sheet is **not** handed in with your Question/Answer Booklet.

See next page

This section has **25** questions. Answer **all** questions on the separate Multiple-choice Answer Sheet provided. For each question shade the box to indicate your answer. Use only a blue or black pen to shade the boxes. If you make a mistake, place a cross through that square then shade your new answer. Do not erase or use correction fluid/tape. Marks will not be deducted for incorrect answers. No marks will be given if more than one answer is completed for any question.

Suggested working time: 45 minutes.

1. An element, X, was found to have the following successive ionisation energies:

Ionisation	1st	2nd	3rd	4th	5th	6th	7th	8 th
Ionisation energy/kJ mol ⁻¹	736	1450	7740	10500	13600	18000	21700	25600

Which of the following represents the **most** likely formula for the sulphate of X?

- (a) XSO₄
- (b) X₂SO₄
- (c) X₂(SO₄)₃
- (d) X(SO₄)₂

2. Which of the following bonds would be expected to have the **greatest** polarity (i.e. most polar?)

- (a) N–F
- (b) P–F
- (c) S–F
- (d) O–F

3. Carbon, silicon, germanium, tin, and lead are all placed together in the same group of the periodic table because:

- (a) They have the same number of valence electrons.
- (b) Each has a structural form with delocalized electrons.
- (c) Atoms of each element have the same number of electrons.
- (d) There is a gradual increase in their atomic masses.

4. Arsenic has 23 isotopes, the lightest of which is ⁶⁵As. Which of the following correctly lists the number of subatomic particles in the arsenide (⁶⁵As³⁻) ion?

- (a) 65 protons, 33 neutrons and 68 electrons
- (b) 33 protons, 32 neutrons and 30 electrons
- (c) 65 protons, 32 neutrons and 65 electrons
- (d) 33 protons, 32 neutrons and 36 electrons

5. Place the following substances in order of melting point, starting with the **lowest**:

Na₂O(s), SO₂(s), O₂(s), SiO₂(s)

- | | | | | |
|-----|---------------------|---------------------|----------------------|----------------------|
| (a) | SO ₂ (s) | O ₂ (s) | Na ₂ O(s) | SiO ₂ (s) |
| (b) | SO ₂ (s) | O ₂ (s) | SiO ₂ (s) | Na ₂ O(s) |
| (c) | O ₂ (s) | SO ₂ (s) | Na ₂ O(s) | SiO ₂ (s) |
| (d) | O ₂ (s) | SO ₂ (s) | SiO ₂ (s) | Na ₂ O(s) |

6. Which of the following statements about carbon dioxide, nitrogen dioxide, and sulfur dioxide is /are **true**?

- (i) They are all covalently bonded molecules
 - (ii) They all have the same total number of electrons
 - (iii) They all have double bonds
 - (iv) They all have the same number of bonding pairs of electrons
-
- (a) (i) and (iii) only
 - (b) (ii) and (iv) only
 - (c) (i), (iii) and (iv) only
 - (d) (i), (ii), (iii) and (iv)

7. Which one of the following molecules has a linear shape?

- (a) CH₄
- (b) H₂Se
- (c) CS₂
- (d) NH₃

8. Predict which of the following has the **greatest** solubility in water.

- (a) propanoic acid
- (b) propan-1-ol
- (c) silver iodide
- (d) magnesium phosphate

9. Predict which one of the following aqueous solutions is the **best** conductor of electricity.

- (a) 1.00 mol L⁻¹ ethanoic acid
- (b) 1.00 mol L⁻¹ ammonia
- (c) 1.00 mol L⁻¹ calcium chloride
- (d) 1.00 mol L⁻¹ nitric acid

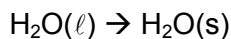
10. The solubility of oxygen in seawater at 20°C is 14.4 mg L⁻¹. Which of the following is the maximum concentration of oxygen in seawater, given in mol L⁻¹, at this temperature?

- (a) 4.5 x 10⁻¹ mol L⁻¹
- (b) 9.0 x 10⁻¹ mol L⁻¹
- (c) 4.5 x 10⁻⁴ mol L⁻¹
- (d) 9.0 x 10⁻⁴ mol L⁻¹

11. Which of the following contains the **largest** number of atoms?

- (a) 6.022×10^{22} molecules of methane
- (b) 0.24 mol of argon gas
- (c) 12 L of nitrogen gas at standard temperature and pressure (STP)
- (d) 10 g of nitrogen monoxide

12. The sign of ΔH for the process:



- (a) is positive and the reaction is endothermic.
- (b) is positive and the reaction is exothermic.
- (c) is negative and the reaction is endothermic.
- (d) is negative and the reaction is exothermic.

13. Which of the following statements about the effect of a catalyst is **false**?

- (a) A catalyst increases the proportion of particles possessing sufficient energy to react.
- (b) A catalyst increases the energy of reactants, enabling them to collide with enough energy to react.
- (c) A catalyst provides an alternative reaction pathway.
- (d) A catalyst is not consumed in a reaction.

14. Which of the following lists the 1 mol L⁻¹ solutions in of **increasing** pH?

- (a) $\text{CH}_3\text{COOH}(\text{aq})$, $\text{NH}_4\text{Cl}(\text{aq})$, $\text{NaCl}(\text{aq})$, $\text{CH}_3\text{CH}_2\text{NH}_2(\text{aq})$
- (b) $\text{CH}_3\text{COOH}(\text{aq})$, $\text{NaCl}(\text{aq})$, $\text{NH}_4\text{Cl}(\text{aq})$, $\text{CH}_3\text{CH}_2\text{NH}_2(\text{aq})$
- (c) $\text{CH}_3\text{CH}_2\text{NH}_2(\text{aq})$, $\text{NaCl}(\text{aq})$, $\text{NH}_4\text{Cl}(\text{aq})$, $\text{CH}_3\text{COOH}(\text{aq})$
- (d) $\text{NH}_4\text{Cl}(\text{aq})$, $\text{CH}_3\text{COOH}(\text{aq})$, $\text{CH}_3\text{CH}_2\text{NH}_2(\text{aq})$, $\text{NaCl}(\text{aq})$

15. During an experiment to establish the concentration of an oxalic acid solution, a student carried out the following steps.

- Rinsed a burette with distilled water.
- Rinsed a conical flask with distilled water.
- Filled the burette with a standardised sodium hydroxide solution.
- Rinsed a pipette with the solution of oxalic acid.
- Dispensed 25 mL of oxalic acid into the conical flask using the pipette.

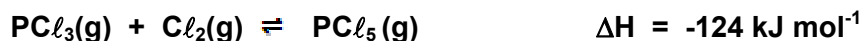
What effect would her procedure have had on the accuracy of her results?

- (a) An artificially large volume of sodium hydroxide is needed to reach the end point, and the calculated concentration of oxalic acid would have been too low.
- (b) An artificially small volume of sodium hydroxide is needed to reach the end point, and the calculated concentration of oxalic acid would have been too low.
- (c) An artificially large volume of sodium hydroxide is needed to reach the end point, and the calculated concentration of oxalic acid would have been too high.
- (d) An artificially small volume of sodium hydroxide is needed to reach the end point, and the calculated concentration of oxalic acid would have been too high.

16. What would be the **most** likely pH of a 0.10 mol L^{-1} solution of sulfuric acid?

- (a) Less than 0.7
- (b) Between 0.7 and 1
- (c) Exactly 1
- (d) Approximately 1.5

The next three questions (17, 18 and 19) refer to the following reaction between phosphorus trichloride and chlorine to form phosphorus pentachloride:



17. If phosphorus trichloride and chlorine were placed in a sealed insulated vessel together with a catalyst, which of the following would **not** cause an increase in the rate at which equilibrium would be attained?

- (a) Increasing the volume of the vessel
- (b) Increasing the temperature
- (c) The addition of chlorine to the reaction mixture
- (d) Increasing the state of sub-division of the catalyst

18. Which of the following statements is/are **true** when the system is at equilibrium?

- (i) Reactants are no longer turning into products.
 - (ii) The concentration of chlorine in the vessel is constant.
 - (iii) Adding a catalyst would not affect the proportions of the reaction mixture.
- (a) (iii) only
 - (b) (i) and (ii) only
 - (c) (ii) and (iii) only
 - (d) (i), (ii) and (iii)

19. Which of the following changes would **not** be observed once equilibrium is re-established if a little phosphorus trichloride gas was added at constant temperature to an equilibrium mixture containing the three gases in the equation?

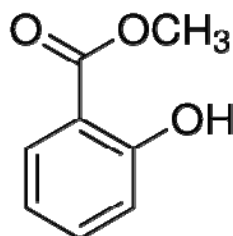
- (a) The reverse reaction would speed up.
- (b) The concentration of phosphorus trichloride would increase.
- (c) The mass of chlorine in the vessel would fall.
- (d) The value of the equilibrium constant, K , would fall.

20. Which of the following equations does **not** represent a redox reaction?

- (a) $\text{Mg}(\text{s}) + 2 \text{HCl}(\text{aq}) \rightarrow \text{MgCl}_2(\text{aq}) + \text{H}_2(\text{g})$
- (b) $\text{Mg}(\text{OH})_2(\text{s}) + 2 \text{CH}_3\text{COOH}(\text{aq}) \rightarrow \text{Mg}(\text{CH}_3\text{COO})_2(\text{aq}) + 2 \text{H}_2\text{O}(\text{l})$
- (c) $\text{Sn}^{2+}(\text{aq}) + \text{Zn}(\text{s}) \rightarrow \text{Sn}(\text{s}) + \text{Zn}^{2+}(\text{aq})$
- (d) $\text{CH}_4(\text{g}) + 2 \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2 \text{H}_2\text{O}(\text{g})$

21. A piece of cobalt is placed into a solution of copper (II) sulfate. Which of the following statements is **false**?
- (a) Copper is precipitated.
 - (b) There is no change in the number of charged particles in solution.
 - (c) Cobalt dissolves.
 - (d) Copper ions are oxidized.
22. Which of the following metals, when placed in a solution of nickel (II) sulfate, would cause the solution to fade from green to colourless?
- (a) Lead
 - (b) Zinc
 - (c) Chromium
 - (d) Iron

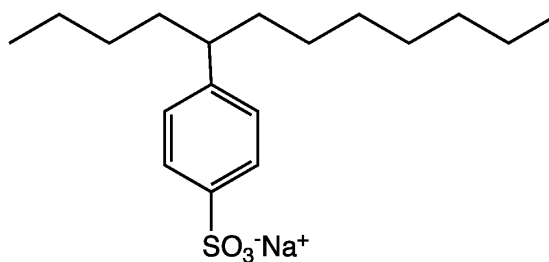
23. Consider the following molecule, commonly known as oil of wintergreen:



oil of wintergreen

- Which of the following statements about oil of wintergreen is **false**?
- (a) It can be synthesised using methanol as one of the starting materials.
 - (b) It is an aldehyde.
 - (c) It is unsaturated.
 - (d) It is an ester.
24. Which of the following molecules can be classified as secondary alcohols?
- (i) 2-methylpropan-2-ol
 - (ii) 3-methylbutan-2-ol
 - (iii) CH₃CHO
 - (iv) CH₃COCH₃
- (a) (i) and (ii) only
 - (b) (ii) only
 - (c) (ii), (iii) and (iv) only
 - (d) (ii) and (iv) only

25. The following formula represents the structure of a compound commonly used as a detergent:



Which of the following statements about detergents is **false**?

- (a) They are often preferred to soaps because of their low tendency to form scum with hard water.
- (b) Part of the molecule is highly water-soluble.
- (c) They are the sodium salts of carboxylic acids.
- (d) The molecules contain large non-polar regions that can form intermolecular forces with fats.

End of section one

See next page